Expected-Information-Gain-Guided Bayesian Calibration of Magnetic-Core Loss and Permeability Models for Power Converters

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Abstract

Magnetic-core models used in power-converter design are commonly fitted to nominal material curves despite limited operating-point coverage and material variation. We present a Bayesian framework that calibrates Steinmetz core-loss and Cole–Cole complex-permeability models and uses expected information gain (EIG) to select measurements. Across five matched-model synthetic data sets, per-parameter median absolute relative errors of the posterior medians ranged from 0.27% to 5.85%, and the known values fell within 28/30 equal-tailed 90% model-conditional posterior credible intervals. In a separate 30-seed paired acquisition study, raw EIG reached a prespecified two-target latent mean-response precision gate using 4–5 measurements; a fixed channel-balanced grid traversal required 9–9, giving a mean paired reduction of 44.8%. The paired count-difference bootstrap 95% interval was 4.00–4.10 measurements, with 100.0% paired wins and a 0.0% raw-EIG gate-failure rate. For accepted measured records, in-sample relative root-mean-square errors ranged from 8.79% to 18.21% for core loss, from 6.89% to 9.33% for μ', and from 36.77% to 52.42% for μ''. The larger μ'' residuals identify a limitation of the tested single-pole permeability model. These results support a reproducible model-conditional calibration workflow and measurement planning within the tested models and operating ranges; they do not establish an optimal laboratory plan.

Keywords: Bayesian calibration; magnetic core; core loss; complex permeability; sequential experimental design; expected information gain.

I. Introduction

Power-converter efficiency, thermal margin, and magnetic-component sizing depend on models for volumetric core loss and complex permeability. Vendor curves provide essential design information, but cover a finite set of frequencies, flux densities, and temperatures and normally do not provide a joint parameter distribution. Fitting one nominal curve can therefore hide parameter non-identifiability and model discrepancy.

Bayesian calibration represents uncertainty in model parameters conditional on the selected model, observations, and prior. Bayesian experimental design ranks a candidate measurement by its expected reduction of posterior uncertainty [1, 2]. Here this framework is implemented as greedy one-step EIG over a finite library, not a globally optimal sequential policy. This is useful when characterization time is limited and dense uniform sweeps are costly. It also makes the measurement objective explicit: the selected point should reduce uncertainty in the quantities represented by the model, not merely occupy an unmeasured location in a frequency–flux grid.

This work contributes (1) a reproducible posterior formulation for Steinmetz and Cole–Cole parameters, (2) a Fisher-spectrum diagnosis of locally constrained parameter combinations, and (3) a paired comparison of EIG acquisition with a fixed channel-balanced grid traversal. The full current version additionally exposes the estimator-convergence campaign, paired outcome construction, modeled-cost endpoint, measured-data acceptance gates, and the acquisition raw-to-aggregate evidence chain. These elements address distinct questions: matched-model recovery checks inference against known synthetic truth, Fisher analysis assesses local identifiability under the specified experiment, acquisition experiments compare policies under common outcomes, and measured-data residuals assess model adequacy. Success in any one branch does not establish the others.

The remainder of the paper reviews magnetic and Bayesian design literature, defines the forward and statistical models, documents the reproducible protocol, and then reports synthetic and measured-data results from one prespecified evaluation.

II. Research Questions and Evidence Scope

The study is organized around four research questions. *RQ1* asks whether the implemented six-coordinate inference problem is locally sensitive in all active directions and whether the posterior computation can recover known matched-model parameters. *RQ2* asks whether greedy raw EIG reaches one declared local precision gate with fewer measurements than the fixed traversal when both policies see identical candidate outcomes. *RQ3* asks the corresponding question for EIG per prespecified modeled cost, evaluated by total modeled acquisition cost rather than count. *RQ4* asks how well the low-order forward laws describe accepted public measured records in sample.

The evidence ladder in Fig. 1 prevents answers from being combined beyond their design. Local Fisher rank is not global identifiability; matched-model recovery is not validation on real ferrites; posterior precision is not truth proximity; and in-sample measured residuals are not held-out predictive accuracy. The acquisition result is therefore an algorithmic, model-conditional comparison under a finite library and a local gate. This explicit estimand is narrower than "optimal magnetic-component identification," but it is reproducible and falsifiable.

The current frozen evidence reports the three-policy comparison that was run: raw EIG, EIG divided by modeled cost, and a deterministic fixed channel-balanced traversal. Randomized balanced, predictive-variance, and Laplace/D-optimal comparators are implemented in the subsequent protocol but have no frozen numerical result in this paper. They are consequently discussed as planned extensions, not

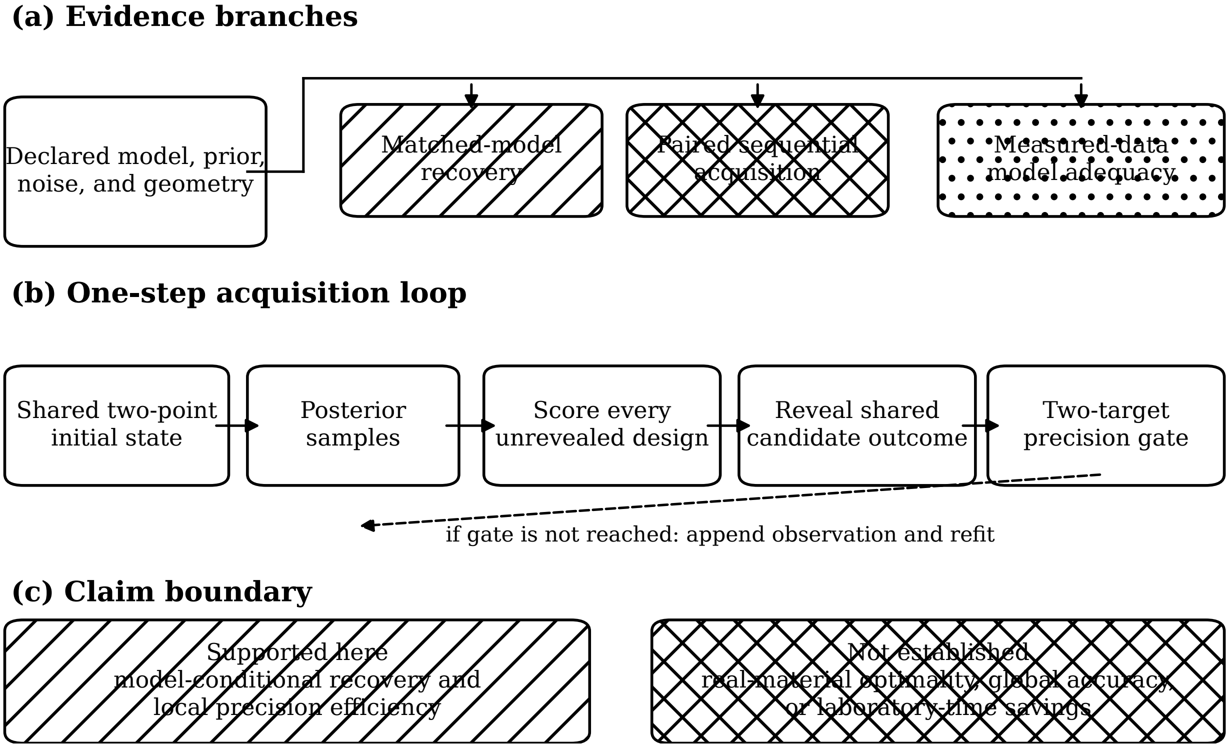

Fig. 1. Study architecture and interpretation boundary. The three evidence branches share declared model and data contracts but answer different questions. In the sequential branch, every policy begins from the same two observations and reveals a pre-generated candidate-indexed outcome. The dashed return arrow denotes refitting when the local precision gate is not reached. Hatch patterns and line styles are used so the figure remains interpretable in grayscale.

as observed outcomes.

III. Related Work

3.1 Magnetic loss and permeability models

The Steinmetz power law remains a compact engineering description for sinusoidal loss, while loss-separation and hysteresis models provide more physical decompositions [3, 4, 5]. Converter waveforms motivated modified and generalized Steinmetz relations that integrate the instantaneous flux-density trajectory or use waveform correction factors [6, 7, 8]. Measurement-led and loss-separation approaches further show why coefficients fitted under one waveform or operating window need not transfer unchanged to another [9, 10]. Subsequent work improved practical loss calculation and reviewed the physical contributions to loss in soft ferrites [11, 12]. Textbook treatments emphasize that material law, geometry, winding, and thermal assumptions must remain distinct in component design [13, 14].

Frequency-dispersive permeability is commonly described through relaxation or resonance spectra. The Cole–Cole law introduced a broadened relaxation distribution [15]; applying its algebraic form to permeability is an engineering adaptation rather than a claim that ferrite magnetization is a dielectric process. Classical ferrite dispersion theory and measurements of Mn–Zn and Ni–Zn ferrites show that domain-wall motion, rotational processes, resonance, and relaxation can all shape μ' and μ'' [16, 17]. The single-pole model used here is therefore intentionally diagnostic: a large loss-component residual is evidence for missing dynamics, not a parameter-estimation failure alone.

Open magnetic datasets now permit reproducible comparisons beyond individual vendor plots. MagNet provides systematically measured core-loss data across materials and excitations [18], while the LEA material database publishes machine-readable permeability and loss records with source metadata [19]. Public data improve traceability but do not substitute for a controlled multi-lot characterization campaign.

3.2 Bayesian calibration and experimental design

Bayesian calibration represents parameter uncertainty conditional on a forward model, likelihood, and prior [20]. Computer-experiment literature distinguishes emulator uncertainty, observation error, calibration parameters, and structural discrepancy [21, 22]. Validation frameworks consequently compare predictions with observations rather than treating posterior concentration as proof of physical validity [23]. Ignoring discrepancy can bias inferred physical parameters and produce overconfident extrapolation [24]. Calibration theory for imperfect simulators also demonstrates that the physical parameter and discrepancy can be confounded unless additional structure is imposed [25, 26]. We do not introduce an unidentified discrepancy process into the present six-parameter fit; instead, measured residuals are reported as a separate adequacy diagnostic.

Bayesian experimental design formalizes the value of a proposed observation through expected utility. Lindley's information criterion [27], foundational reviews [1, 2], and classical optimum-design methods [28] motivate the EIG criterion used here. Estimation remains challenging because the predictive density is generally unavailable; relevant strategies include Laplace and MCMC approximations [29], nested Monte Carlo [30], multilevel estimators [31], stochastic simulation-based optimization [32], and variational bounds [33]. Adaptive-policy amortization offers a scalable direction for longer experimental sequences [34], but the present work deliberately uses a transparent one-step ranking over a finite candidate library.

Information-based acquisition connects Shannon entropy and Kullback–Leibler divergence to active data selection [35, 36, 37]. Classical D-optimal design instead optimizes a determinant of an information matrix and is linked to equivalence theorems [38, 28]. These criteria coincide only under particular approximations; the fixed traversal used in the reported benchmark is neither of them.

IV. MAGNETIC-CORE MODELS

4.1 Core loss

For sinusoidal excitation within a specified operating window, the Steinmetz model is

$$P_v(f, B_{\mathrm{pk}}) = k f^{\alpha} B_{\mathrm{pk}}^{\beta}, \tag{1}$$

where P_v is volumetric loss and (k, α, β) are inferred parameters [3]. The units of k depend on the units used for P_v, f, and B_{pk}. Equation (1) is not temperature dependent in the present implementation. Measured-data analysis therefore uses a specified temperature subset; extrapolation across temperature is out of scope. More general waveform-dependent loss equations remain important for converter application studies [6, 11].

4.2 Complex permeability

We adopt the $e^{j2\pi ft}$ time convention and write $\mu_r = \mu' - j\mu''$, so the reported loss component is $\mu'' = -\operatorname{Im}\mu_r \geq 0$ for a passive response. The relative complex permeability is represented with a single Cole–Cole relaxation [15],

$$\mu_r(f) = 1 + \frac{\mu_s - 1}{1 + (jf/f_{\mathrm{rel}})^{1-a_{\mathrm{cc}}}}, \tag{2}$$

where μ_s is low-frequency permeability, f_{rel} is a relaxation frequency, and a_{cc} broadens the relaxation. Storage and loss components are evaluated separately because a model can fit μ' while failing to represent μ''. A one-pole model is a deliberate low-order approximation, not a complete ferrite loss model.

Equation (2) fixes the high-frequency asymptote to unity and absorbs the angular-frequency convention into f_{rel}. The real and loss components follow from the same complex denominator, which prevents independent tuning of μ' and μ''. For an optional inductance channel, the real component is mapped through $L_m = \mu_0 \mu' N^2 A_e / \ell_e$ using specified geometry [13, 14]. Geometry is never inferred jointly with material parameters in the present study.

4.3 Validity window

Both forward laws are isothermal and evaluated only inside specified data ranges. A data point is defined by channel, frequency, peak flux density, and temperature. Core-loss points require positive B_{pk}; permeability points use small-signal records. The likelihood rejects mixed temperature cohorts. Measured permeability uses material-specific cohorts of $25 \pm 0.75\,^\circ$C for N87 and $30 \pm 0.75\,^\circ$C for N95. Measured core loss uses $25 \pm 0.5\,^\circ$C for N49, N87, and N95, and $100 \pm 0.5\,^\circ$C for 3C95. These restrictions prevent an apparently precise posterior from silently extrapolating across temperature, waveform, or excitation regime.

V. BAYESIAN CALIBRATION AND DESIGN

Let $\theta = (\log k, \alpha, \beta, \log \mu_s, \log f_{\mathrm{rel}}, a_{\mathrm{cc}})$ denote the active transformed parameters. Log coordinates enforce positivity of scale parameters while retaining direct coordinates for the exponents. For observation y_o at design d_o, the likelihood is

$$p(\mathbf{y} \mid \theta, \mathbf{d}) = \prod_o \mathcal{N}\left(y_o; F_o(\theta, d_o), \sigma_o^2\right), \tag{3}$$

with fixed relative scales. Synthetic scales are $3\mu'$, 5relative scale as a weighting model. The posterior is

$$p(\theta \mid \mathbf{y}, \mathbf{d}) \propto p(\mathbf{y} \mid \theta, \mathbf{d}) p(\theta). \tag{4}$$

Priors are defined directly in the transformed space used by the sampler, so no inverse-transform Jacobian is added to this active-coordinate density. Bounds encode physically admissible numerical domains, not empirical proof that the chosen model is correct.

The six coordinates divide into two response blocks in the likelihood but are sampled jointly. Core-loss observations directly update $(\log k, \alpha, \beta)$, while permeability and inductance observations update $(\log \mu_s, \log f_{\mathrm{rel}}, a_{\mathrm{cc}})$. Joint sampling retains a single posterior contract and permits one acquisition library to contain all channels; it does not create physical coupling absent from the forward laws. Table I makes this structural separation explicit.

Four uncertainty sources must be distinguished. *Parameter uncertainty* is represented by posterior draws conditional on the chosen model and fixed geometry. *Observation uncertainty* enters through the declared Gaussian scale. *Monte Carlo uncertainty* arises because posterior and EIG integrals are approximated by finite samples. *Structural uncertainty* is exposed by measured residuals but is not parameterized in the likelihood. Only the first three are quantified in parts of the present workflow; an EIG score interval is therefore not a total physical-uncertainty interval.

5.1 Posterior computation and diagnostics

Posterior sampling uses the affine-invariant ensemble construction [39] implemented by `emcee` [40]. Each ensemble retains 5000 steps after 1000 warm-up steps for each of 48 walkers. Initial states are perturbations of the prior center and both initialization and proposal streams are deterministically seeded. A fit is accepted only with finite integrated autocorrelation estimates, at least 50 retained steps per autocorrelation time, effective sample size at least 400, mean acceptance between 0.20 and 0.60, and finite log probability for every retained sample.

These checks follow the principle of comparing Monte Carlo variation with between-sequence behavior [41] and modern effective-sample-size practice [42], but interacting ensemble walkers are not treated as independent chains. Accordingly, the paper does not report a conventional $\widehat{R}$ for

TABLE I: ACTIVE PARAMETERS, ROLES, TRANSFORMATIONS, AND PRINCIPAL CHANNELS.

Physical parameter	Active coordinate	Role	Principal response	Main limitation
k	$\log k$	core-loss scale	P_v	units follow the declared convention
α	α	frequency exponent	P_v	local to sinusoidal, isothermal data
β	β	flux-density exponent	P_v	no waveform or DC-bias dependence
μ_s	$\log \mu_s$	low-frequency permeability	μ', L_m	fixed rather than inferred geometry
f_{rel}	$\log f_{\mathrm{rel}}$	relaxation frequency	μ', μ''	one effective relaxation only
a_{cc}	a_{cc}	relaxation broadening	μ', μ''	no multiple resonances

the ensemble. Diagnostic failure invalidates a measured fit for quantitative publication rather than being repaired by discarding an unfavorable seed.

Matched-model recovery also exercises the complete forward–likelihood– sampler–summary path against known generating values. It is an implementation check related to posterior-quantile validation [43], but five independent fits do not constitute simulation-based calibration [44]. A calibrated rank experiment would require many more prior-predictive replications and a uniform-rank diagnostic. Posterior and residual graphics are interpreted as part of a Bayesian workflow rather than as substitutes for numerical diagnostics [45, 46].

The local Fisher approximation [47] is

$$\mathscr{I}_{ij} = \sum_o \frac{1}{\sigma_o^2} \frac{\partial F_o}{\partial \theta_i} \frac{\partial F_o}{\partial \theta_j}. \tag{5}$$

Its spectrum is computed over the exact active vector and with reported scaling. A large condition number indicates locally sloppy combinations but does not by itself establish global non-identifiability [48]. Geometric analyses of sloppy models likewise interpret broad eigenvalue spectra as anisotropic distinguishability, not as a universal parameter ranking [49]. The spectrum is therefore reported together with its dimension and scaling convention.

Let $\mathscr{D}_t = (\mathbf{y}_t, \mathbf{d}_t)$ denote the observations available before acquisition step t. For candidate d, the one-step EIG is the conditional mutual information [27, 1]

$$\mathrm{EIG}_t(d) = \mathbb{E}_{\substack{\theta \sim p(\theta \mid \mathscr{D}_t) \\ y \sim p(y \mid \theta, d)}} \left[\log p(y \mid \theta, d) - \log p(y \mid d, \mathscr{D}_t)\right], \tag{6}$$

where

$$p(y \mid d, \mathscr{D}_t) = \int p(y \mid \vartheta, d) p(\vartheta \mid \mathscr{D}_t)\, d\vartheta. \tag{7}$$

For outer draws (θ_i, y_i) and inner posterior draws ϑ_{ij}, the implemented nested estimator is

$$\widehat{U}(d) = \frac{1}{N} \sum_{i=1}^{N} \left[\log p(y_i \mid \theta_i, d) - \log\left\{ \frac{1}{M} \sum_{j=1}^{M} p(y_i \mid \vartheta_{ij}, d) \right\} \right]. \tag{8}$$

Finite inner budgets introduce bias through the logarithm and finite outer budgets introduce variance; close candidates can therefore exchange rank [30, 31]. The convergence campaign described below tests ranking and downstream endpoint stability rather than assuming one budget is sufficient.

The reported estimator draws 1200 outer parameter–observation pairs and uses 400 inner posterior draws to approximate the posterior-predictive density [29, 30]. Each score is repeated 40 times; the mean, standard error, 95% Monte Carlo interval, and top-rank frequency are retained. Candidate streams are seeded from a stable key comprising channel, frequency, flux density, and temperature, so list permutation cannot change a design's draws. We evaluate two distinct objectives: raw EIG for the measurement-count endpoint and EIG divided by a prespecified modeled acquisition cost for the total-cost endpoint. These intervals quantify repeated nested-Monte-Carlo variability conditional on the retained posterior draws and model; they exclude chain-to-chain, structural-model, geometry, and noise-model uncertainty.

5.2 Paired sequential policy

After the same two initial observations—P_v at 30 kHz, 0.05 T, and 25 °C, and μ' at 10 kHz and 25 °C—each policy selects one unrevealed candidate, receives its pre-generated noisy outcome, refits the posterior, and checks the common precision gate. Raw EIG maximizes estimated information, EIG/cost maximizes information per modeled cost, and the comparator follows a deterministic fixed channel-balanced grid traversal. It is not uniform random sampling, a space-filling design, or a D-optimal design. All policies share the candidate library, latent truth, candidate-indexed noisy outcomes, maximum budget, and stopping rule. The candidate library is isothermal, so it contains no temperature-only duplicates of the implemented forward laws. The numerical settings are summarized in Table III.

The stopping statistic is the half-width of the central 90% posterior interval for the noise-free latent mean response divided by the absolute posterior median. Observation noise is not added to this gate. The two fixed targets are $P_v(100\,\mathrm{kHz}, 0.1\,\mathrm{T}, 25\,^\circ\mathrm{C})$ at 8% and $L_m(100\,\mathrm{kHz}, 25\,^\circ\mathrm{C})$ at 5%. This is a local precision criterion: it checks neither truth proximity, six-parameter recovery, other channels, held-out performance, nor model adequacy. A policy that misses either threshold is recorded as a failure at the maximum budget. Candidate EIG integrates over the joint six-parameter posterior; only the stopping and evaluation endpoint is restricted to these two responses.

VI. COMPUTATIONAL REPRODUCIBILITY

All experiments use fixed software versions, specified model and inference settings, explicit data selections, and deterministic random seeds. Results are retained at the individual-seed level and are accepted for aggregation only when all prespecified validity gates pass.

TABLE II: FINITE SYNTHETIC ACQUISITION LIBRARY AND DECLARED CHANNEL ASSUMPTIONS.

Channel	Frequencies (kHz)	B_{pk} (T)	Points	Relative noise	Modeled cost
P_v	30, 50, 100, 200, 300, 500	0.05, 0.10, 0.20	18	3%	60
μ'	10, 30, 100, 300, 500, 1000, 2000, 3000	0	8	2%	20
μ''	10, 30, 100, 300, 500, 1000, 2000, 3000	0	8	5%	20
L_m	10, 100, 1000	0	3	1%	15

All 37 candidates are at 25 °C. Modeled costs are positive prespecified units used only for the EIG/cost endpoint; they are not measured instrument or laboratory durations. The two initial observations belong to the same library and count toward the stopping total.

TABLE III: COMPUTATIONAL SETTINGS USED FOR THE REPORTED EXPERIMENTS.

Item	Setting
Ensemble sampler	48 walkers; 1000 warm-up + 5000 retained
Convergence criteria	ESS $\geq$ 400; steps/$\tau \geq$ 50
Acceptance criterion	0.20–0.60
Nested EIG	1200 outer; 400 inner; 40 score replicates
Sequential budget	at most 25 measurements
Acquisition endpoints	raw EIG/count; EIG/cost/modeled cost
Temperature cohorts	material and channel specific, as stated in text
Precision gate	latent mean response: 8% P_v; 5% L_m

The analysis checks all required seed and material cases before forming aggregate summaries. Records with nonfinite values, failed convergence, or boundary-active parameters are excluded according to criteria specified before evaluation. Numerical statements, tables, and figures are generated programmatically from the same complete set of accepted records, avoiding manual transcription or selective replacement of outcomes.

6.1 Estimator-budget qualification

The nested estimator was qualified before the 30-seed acquisition campaign. Twelve posterior states were formed from four seeds (7100–7103) after 2, 4, and 6 observations. At each state, 27 candidate settings crossed outer budgets $\{100, 300, 900\}$, inner budgets $\{50, 100, 300\}$, and replicate counts $\{5, 10, 20\}$. Every setting was compared with a 1200-outer, 400-inner, 40-replicate reference. Two sentinel states were additionally recomputed at 2400 outer and 800 inner draws with 40 replicates. Prefix-nested random streams made the lower-budget samples literal prefixes of the larger calculations, reducing irrelevant differences between settings.

The acceptance decision combined the stable top candidate, Spearman rank correlation, score-interval overlap, relative reference regret, and exact agreement of downstream count and modeled-cost endpoints. Ten disjoint seeds (7200–7209) supplied that downstream check. The selected setting was then locked before the confirmatory acquisition seeds 7300–7329 were evaluated. This separation prevents selecting an estimator budget because it happened to produce the most favorable headline comparison.

The public audit projection retains the campaign records needed to reconstruct the acquisition endpoints and estimator decision while removing machine paths and scheduler metadata. Posterior arrays are retained flattened over sampling draws; because walker and iteration indices are not present, they are not described as raw MCMC chains. The compact frozen summary authenticates the reported measured-fit aggregates and exclusions, but the audit projection does not independently reconstruct those fits. Raw upstream measured curves remain governed by their source repositories and are not redistributed here.

VII. EVALUATION PROTOCOL

Five fixed seeds are used for matched-model recovery. A disjoint, prespecified set of 30 seeds is used for the paired acquisition comparison only after the nested-Monte-Carlo setting passes a separate convergence study. That study uses prefix-nested common random numbers and a deterministic paired bootstrap of the replicate-mean selector; it requires stable candidate and reference winners, rank correlation, score-interval overlap, low reference regret, and downstream endpoint agreement. The policies share initial observations, candidate library, realized candidate outcomes, noise policy, budget, and stopping rule. Results include per-seed counts, modeled costs, and failures, not only an aggregate ratio. The stopping rule concerns posterior precision of two latent mean responses and is not an accuracy or coverage guarantee.

For recovery, observations are generated from the same six-parameter forward family used in inference. A parameter vector is drawn independently for each seed from a datasheet prior fixed before that draw; the prior is never rebuilt from the realized truth. Sampler walkers are initialized around the fixed prior center, not around the hidden generating vector. Observation noise is then generated independently. This is a prior-predictive matched-model design: it removes oracle centering while retaining favorable equality between the data-generating and inference model families. The reported error is the absolute relative difference between posterior median and the generating value. The interval statistic counts whether that value falls within the equal-tailed 90% model-conditional posterior credible interval. With only five seeds, the count is a descriptive implementation check, not a coverage experiment. The Fisher matrix is evaluated in the exact transformed coordinate system used by sampling.

For the paired acquisition analysis, a stable design-keyed seed generates one noisy observation for every candidate before any policy is run. A policy reveals the already generated value when it selects that design. Thus policy order cannot change the noise attached to a candidate, and all policies in a seed share the same latent truth and candidate outcome map. The 30 paired differences are the unit of analysis. Descriptive means, medians, standard deviations, win and failure rates, and percentile intervals are computed from those differences. The reported confidence interval uses 10,000 deterministic paired bootstrap resamples. It quantifies between-seed variation in this matched-model experiment and is not a population-wide material claim.

TABLE IV: EVIDENCE LAYERS RETAINED FOR THE CURRENT FROZEN RELEASE.

Layer	Prespecified cases	Retained public evidence	Scientific role
Matched recovery	5 generating seeds	seed-level parameter summaries	implementation check
Estimator states	$4 \times 3 = 12$	state records and flattened posterior samples	budget qualification
Estimator grid	27 budget combinations across 12 states	108 candidate-score records plus 12 references	ranking stability
Doubled-budget audit	2 sentinel states	2 doubled-budget score records	reference sensitivity
Downstream validation	10 seeds	10 policy endpoint records	endpoint stability
Confirmatory acquisition	30 paired seeds	30 complete policy trajectories	count and cost endpoints
Measured adequacy	declared material/source pairs	aggregate metrics and disclosed exclusions	model discrepancy diagnostic

TABLE V: MATCHED-MODEL RECOVERY AND INCLUSION OF THE GENERATING VALUE IN EQUAL-TAILED 90% POSTERIOR CREDIBLE INTERVALS (FIVE SYNTHETIC SEEDS).

Parameter	Median error (%)	Truth included
k	5.85	4/5
α	0.27	4/5
β	0.36	5/5
μ_s	0.38	5/5
f_{rel}	0.87	5/5
a_{cc}	0.62	5/5

The measured evaluation uses public complex-permeability curves for specified N95, N87, and 3C95 source/material pairs and datasheet core-loss curves for N49, N87, N95, and 3C95 [19]. Input-data versions, temperature cohorts, unit conversions, and range filters were fixed before inference. Storage/loss permeability errors, core-loss residuals, boundary activity, and operating-window coverage are reported separately. Records with failed convergence or active boundary flags are retained for diagnostic review but excluded from aggregate numerical claims.

Core-loss adequacy is summarized by in-sample relative root-mean-square error (RRMSE) within each temperature cohort. Permeability adequacy is evaluated separately for storage and loss components because combining them can conceal a poor μ'' fit behind the larger magnitude of μ'. These metrics assess the selected low-order model on observed records; they are neither held-out prediction errors nor component-level thermal validation.

VIII. RESULTS

8.1 Identifiability and synthetic recovery

The synthetic 6×6 Fisher matrix was full rank. As shown in Fig. 2(a), its eigenvalues span 1.24×10^2 to 2.91×10^6, giving a full-matrix condition number of 2.35×10^4. The broad spectrum indicates strong local sensitivity anisotropy in the active transformed coordinates; it does not establish global identifiability. Fig. 2(b) shows the absolute posterior-median errors for each parameter and seed. Across five prior-predictive matched-model data sets, the per-parameter medians range from 0.27% to 5.85%. The known values fell within 28/30 equal-tailed 90% model-conditional posterior credible intervals. This observed inclusion count is descriptive, not a coverage estimate. Table V reports the same matched-model recovery check.

8.2 Nested-estimator qualification

The conservative reference setting of 1200 outer draws, 400 inner draws, and 40 score replicates was selected by the prespecified convergence decision. The doubled-budget sentinels preserved the required selector behavior, and the ten downstream validation seeds preserved both count and modeled-cost endpoints. This result qualifies the numerical setting for the subsequent campaign; it does not make each finite EIG score exact. In particular, the reported score intervals remain conditional on the retained posterior and forward model.

8.3 Paired synthetic acquisition

Fig. 3 makes the paired outcome structure explicit. Raw EIG reached the two-target latent mean-response precision gate with 4–5 measurements, while the fixed channel-balanced grid traversal required 9–9. Raw EIG used fewer measurements in every paired run, with a mean reduction of 44.8%. These results evaluate selection efficiency for the count endpoint in this 30-seed matched-model experiment. The mean paired count difference has a bootstrap 95% interval of 4.00–4.10 measurements; the paired win rate is 100.0% and the raw-EIG gate-failure rate is 0.0%. The seed-level distribution is concentrated rather than hidden by the mean: 29 pairs favor raw EIG by four measurements and one pair favors it by five. For the separate modeled-cost endpoint, EIG/cost gave a descriptive mean paired reduction of 34.4%; neither policy comparison had a gate failure. The cost units are prespecified rather than measured laboratory time.

8.4 Measured-data model adequacy

Fig. 4 compares the in-sample relative root-mean-square errors (RRMSEs) of the accepted measured-data fits. For LEA Material Database permeability records that satisfied the convergence criteria, RRMSEs were 9.33%/52.42% for μ'/μ'' on N87 and 6.89%/36.77% on N95. The larger μ'' errors expose discrepancy in the one-pole model. N87 and 3C95 MagNet records failed convergence or boundary gates and were excluded from numerical claims [18]. Measured Steinmetz fits produced in-sample RRMSEs of 8.79% (N87), 12.55% (N49), 13.40% (3C95), and 18.21% (N95) within their specified temperature cohorts. These are fit residuals, not held-out errors.

All numerical results in this section, including Figs. 2, 3, and 4, use the same prespecified models, data filters, random seeds, and evaluation criteria.

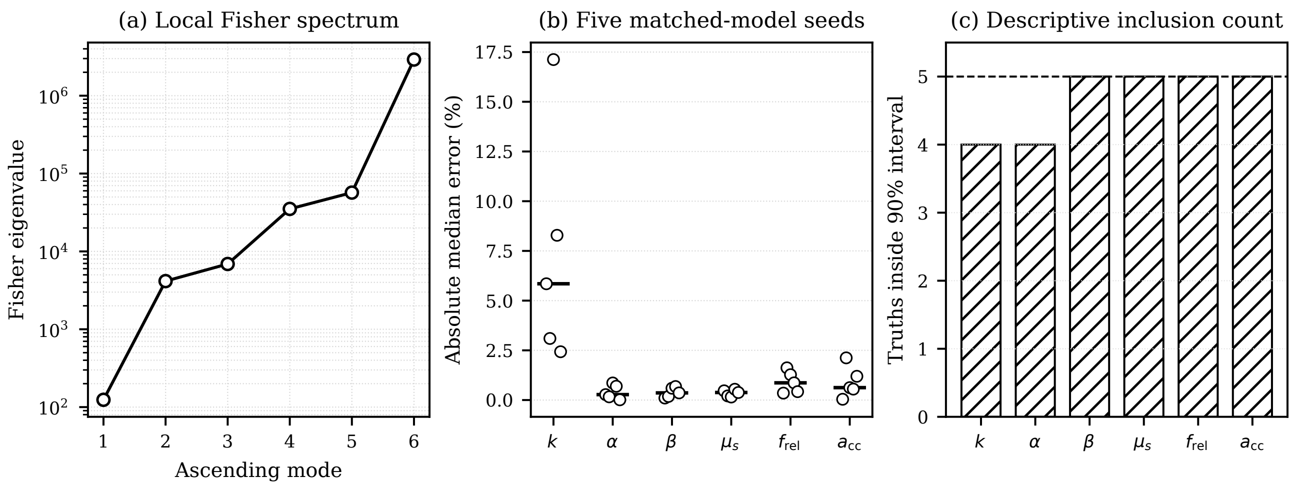

Fig. 2. Synthetic local identifiability and matched-model recovery. (a) Fisher eigenvalues in ascending order. (b) Absolute posterior-median errors across five seeds; circles are horizontally offset for visibility and horizontal ticks mark medians. (c) Number of the five generating values inside each equal-tailed 90% model-conditional posterior interval. The five-seed inclusion count is descriptive and is not a coverage estimate.

IX. Discussion and Limitations

9.1 Interpretation of the evidence

The framework separates whether available measurements constrain parameters within a chosen model from whether that model represents measured material behavior. Fisher information and posterior width address the former locally or conditionally; reported in-sample residuals expose, but do not fully quantify, the latter. Held-out predictive validation remains future work.

The synthetic results support implementation-level conclusions. Full local rank shows that every active coordinate affects the specified synthetic experiment, while the broad spectrum shows that those effects occur at very different scales. The recovery table then checks the nonlinear posterior under matched truth. Neither result implies that the same coordinates are uniquely physical when the material response lies outside the six-parameter family. This distinction matters for converter models because several parameter sets can produce similar loss or permeability over a narrow operating window yet diverge when frequency, flux density, or temperature changes.

The paired acquisition result shows that raw EIG exploited posterior structure in the matched problem: it reached the two-target precision gate with fewer measurements for every reported seed under shared candidate outcomes. The result means only that fewer measurements were needed to narrow two specified latent mean-response intervals. It does not establish accurate recovery of the magnetic component, global predictive accuracy, or robustness to model misspecification.

9.2 Implications for converter modeling

For a converter workflow, the posterior is most useful as a conditional input to loss, inductance, and sensitivity calculations over the specified operating region. Posterior draws can be propagated through a component model instead of substituting one nominal coefficient set, allowing a designer to determine whether parameter uncertainty materially affects thermal or magnetic design margins. The Fisher spectrum can motivate testing whether a new channel or a wider, physically controlled excitation range would constrain a weak direction more effectively than densifying the original sweep [1, 28].

Measured residuals define a separate model-adequacy assessment. The accepted core-loss fits show the error remaining after estimating Steinmetz coefficients inside one temperature cohort. The much larger loss-component permeability residuals show that a single Cole–Cole pole cannot be treated as a broadband physical description for these records. A component calculation that is sensitive to μ'' should therefore retain an explicit discrepancy allowance or adopt a higher-order permeability law before relying on posterior parameter width. Conversely, a reasonable μ' residual does not validate μ'', because the two components occupy different numerical scales and correspond to different design consequences.

The common evaluation protocol safeguards this interpretation by preventing combination of a favorable recovery run, a figure from another seed set, and a measured fit produced under different filters. Fixed model settings, data selections, and random seeds also make it possible to distinguish sampling variation from changes to the computational experiment.

9.3 Threats to validity

Internal validity is limited by five synthetic recovery seeds and 30 paired acquisition seeds. Pairing controls differences in candidate outcomes between policies. A prespecified convergence study compares candidate ranks and downstream endpoints against a higher-budget reference; finite nested Monte Carlo variance nevertheless remains. Synthetic recovery is matched-model, and the observed interval-inclusion count is too small to establish calibrated coverage.

The policy comparison is against one fixed channel-balanced traversal; it does not establish dominance over randomized balanced, predictive-variance, Fisher-greedy, or D-optimal designs. Observation noise and geometry are fixed,

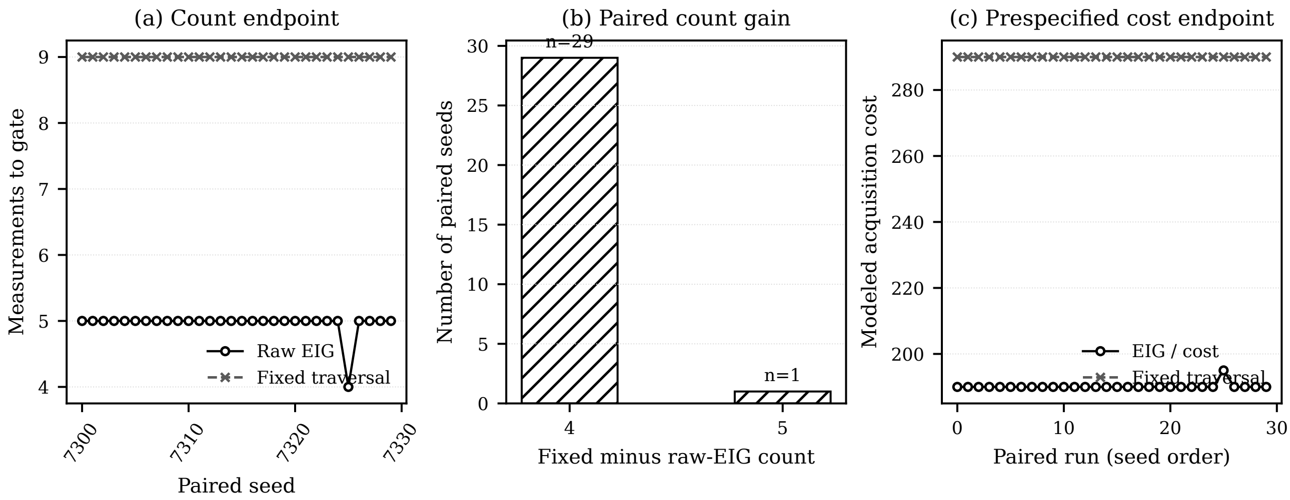

Fig. 3. Thirty-seed paired matched-model acquisition results. (a) Raw-EIG and fixed-traversal measurement counts at the common two-target precision gate. (b) Distribution of paired count gains, defined as fixed minus raw EIG. (c) Total prespecified modeled cost for EIG/cost and the fixed traversal. Candidate-indexed outcomes are identical across policies within each seed; modeled cost is an assumed endpoint, not observed laboratory duration.

and estimator intervals are not total physical-uncertainty intervals.

Construct validity is limited by the low-order forward models and available data. The Steinmetz relation is temperature independent, and one Cole–Cole pole does not capture all ferrite relaxation, resonance, or loss processes [17]. Measured errors are in-sample. Public curves may also contain digitization, instrument, preprocessing, and source-specific errors not represented by the likelihood. Excluding convergence-invalid or boundary-active records prevents unsupported aggregate claims but restricts the population represented by the accepted subset.

External validity is confined to the specified channels, temperature cohorts, excitation ranges, materials, and cost model. The measured-data EIG ranking is a model-conditional acquisition suggestion, not a validated optimal laboratory plan, particularly given the observed μ'' discrepancy. Measurement-count reductions must not be interpreted as laboratory-time savings because setup, settling, calibration, rejection, and batch costs were not measured. The results do not replace held-out validation, controlled multi-lot characterization, switching-waveform experiments, or component qualification. Those extensions require new experimental evidence rather than broader interpretation of the present results.

9.4 Evidence required for broader claims

Three additions would materially change the evidential scope. First, a new checksum-locked paired campaign must compare EIG with randomized balanced, predictive-variance, and Fisher/Laplace D-optimal policies under the same outcome table, count budget, and cost table. The implemented follow-on protocol declares these methods but the current release contains no outcome from it. Second, a disjoint latent-response grid must test truth error and interval inclusion away from both acquisition candidates and stopping targets. Such a test can detect a narrow-but-wrong matched-model posterior, although it still cannot diagnose structural discrepancy. Third, a laboratory study needs independently calibrated noise, measured acquisition durations, multiple lots, and held-out temperature/frequency/flux regions. Only that third layer could support claims about real-material measurement or time savings.

This hierarchy also defines negative results that would revise the conclusion: unstable EIG rankings under increased Monte Carlo budgets, loss against stronger paired comparators, poor disjoint-grid prediction despite reaching the gate, or failure of real-data residual and coverage criteria. Publishing these decision boundaries makes the present positive result testable rather than open ended.

X. Conclusion

Bayesian calibration recovered the six active Steinmetz and Cole–Cole parameters with per-parameter median absolute relative errors of 0.27%–5.85% across five matched-model synthetic data sets. The known values fell within 28/30 equal-tailed 90% model-conditional posterior credible intervals. Under paired candidate outcomes and a common stopping rule, raw EIG reached the two-target latent mean-response precision gate with 4–5 measurements, compared with 9–9 for the fixed channel-balanced grid traversal, a descriptive mean reduction of 44.8% across 30 seeds. The paired count-difference bootstrap interval was 4.00– 4.10 measurements, with 100.0% paired wins and a 0.0% raw-EIG gate-failure rate. The accepted measured records produced substantially larger RRMSEs for μ'' than for μ', showing that posterior concentration cannot compensate for an inadequate permeability law. The workflow links inference, local identifiability, measurement selection, model-adequacy checks, and traceable evidence within the tested models, public data records, and operating ranges.

A Exact Sequential Evaluation Logic

For each confirmatory seed, the synthetic truth is drawn once from the fixed prior. One outcome per exact candidate identity is then generated and stored. Each policy is

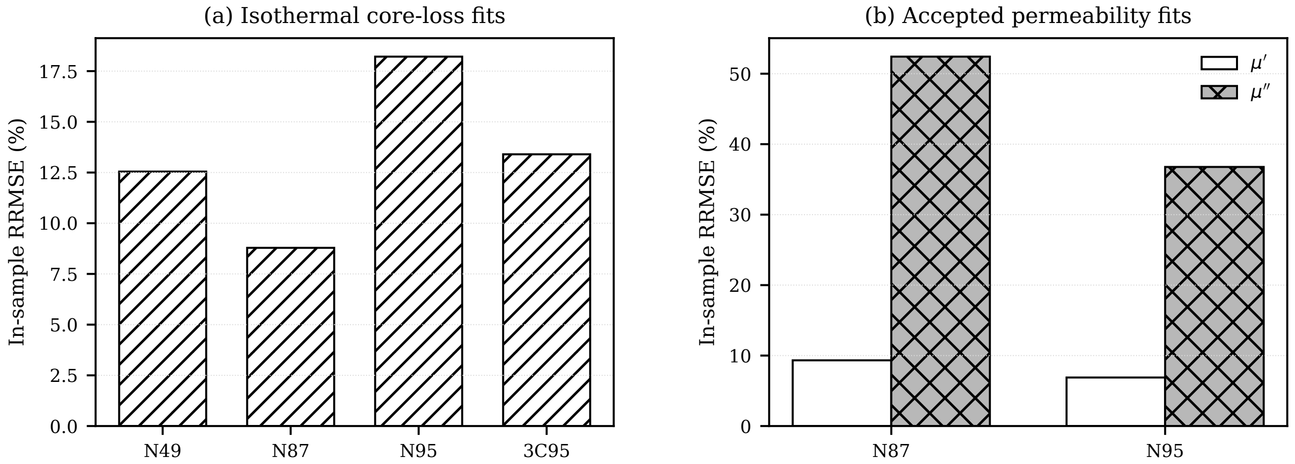

Fig. 4. Measured-data in-sample model adequacy for accepted fits. (a) Isothermal Steinmetz core-loss RRMSE. (b) Separate storage and loss permeability RRMSE for accepted LEA records. The much larger μ'' error is not absorbed into a combined complex-valued metric. Convergence-invalid or boundary-active MagNet fits are excluded from these numerical bars and disclosed in the text.

evaluated by the following logic:

1. Start with the same declared P_v and μ' observations.
2. Fit the six-coordinate posterior and require all sampler gates.
3. If both target intervals satisfy the precision gate, retain count and modeled cost and stop successfully.
4. Otherwise score or traverse only unrevealed candidates. Raw EIG uses $\widehat{U}(d)$; EIG/cost uses $\widehat{U}(d)/c(d)$; the fixed policy uses its deterministic channel-balanced order.
5. Break deterministic ties by exact design identity, reveal the stored outcome for the selected identity, append it, and return to step 2.
6. If the 25-measurement budget is exhausted, retain the trajectory as a failure rather than imputing a stopping count.

This logic separates three random objects: the prior-predictive truth, the candidate-indexed outcome table, and the policy's numerical scoring stream. Changing policy order changes neither the truth nor any candidate outcome. Stable design identities also make EIG scores invariant to permutation of the input library.

B Endpoint and Uncertainty Definitions

For a latent target $g(\theta)$, the reported precision statistic is

$$h_g = 100\,\frac{q_{0.95}[g(\theta)] - q_{0.05}[g(\theta)]}{2\,|q_{0.50}[g(\theta)]|}. \tag{9}$$

The gate requires $h_{P_v} \leq 8$ and $h_{L_m} \leq 5$. The measurement-count endpoint is the number of revealed observations, including the two initial points, at the first satisfied gate. The cost endpoint is the sum of declared channel costs through that same step. Because the cost constants were not measured, the endpoint is dimensionless modeled burden even though the configuration names them as seconds for engineering convenience.

For paired seed s, count gain is $\Delta_s = n_{\text{fixed},s} - n_{\text{rawEIG},s}$; modeled-cost gain is defined analogously for EIG/cost. Resampling is performed over complete paired seeds, never over unpaired policy outcomes. Monte Carlo score intervals come from repeated estimates of Eq. (8); posterior intervals come from retained sampler draws; paired bootstrap intervals come from the 30 seed-level differences. These three intervals answer different questions and are not interchangeable.

C Evidence, Software, and Version Boundary

The current manuscript is tied to frozen release `20260812T035654Z_a0703698ace9` through a machine-readable result lock. Generated tables, macros, figures, and the summary are checksum verified before paper compilation. A separate sanitized audit asset contains the 30 policy trajectories, estimator states and scores, downstream validation records, and retained flattened posterior matrices needed to reconstruct the acquisition and estimator-selection claims. Its scope and digest are recorded in the repository. Raw measured curves are obtained from the cited upstream sources and are not redistributed as if they were produced by this study.

The six-page conference snapshot is an immutable historical record associated with its own earlier release and claim ledger. This full current version keeps the same A4 two-column visual convention but has no page ceiling, includes the 30-seed confirmatory evidence and estimator qualification, and must not be used to retroactively reinterpret the snapshot. Source, protocols, result locks, and verification commands are available in the accompanying public repository.

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